

If Greece Has Recovered, Why Do So Many Households Still Struggle?

Official income poverty affects roughly one person in five. Yet about two households in three report difficulty making ends meet. The distance between them reveals a moving poverty line, an uneven recovery and the continuing weight of unemployment, wages and housing.

1 in 5

19.6%

People at risk of
poverty — the official
measure

2 in 3

66.7%

Households struggling
to make ends meet

The Poverty Rate Says One Thing. Households Say Another.

Looking at the labour market and some headline economic indicators, Greece appears to have recovered from the crisis. Unemployment has fallen sharply. Output has returned. The bailout years are no longer the first fact most economic summaries reach for.

But the closer we move to households, the less complete that recovery looks.

By the official income-poverty measure, Greece is in difficulty, but not exceptional. By what households report, it is almost in a category of its own. That is the paradox this piece unfolds: income poverty and reported hardship are telling two different stories, and the poverty rate becomes misleading when it is read alone, without context.

FIGURE 1 Greece Is Far Above the Poverty-Hardship Line

Does official income poverty account for the hardship Greek households report?

▼ **Methods and limits**

Read with this. Descriptive comparison of two official measures. Reported hardship is answered by HOUSEHOLDS and income poverty counts PEOPLE, so the axis is labelled percent rather than either. In the second view the fitted line excludes Greece, describing the European pattern Greece is being judged against rather than one Greece helped set. The labelled point marks the median country on each measure taken SEPARATELY, so it is a reference point rather than an actual country: no member state necessarily sits there. Both views are country-level and say nothing about any individual household.

How Greece's hardship gap developed										
Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	77.7	76.8	77.2	74.1	71.0	69.8	71.0	68.4	67.0	66.7
EU median: reported hardship	28.9	25.9	22.6	20.5	18.5	16.8	15.4	17.9	16.6	17.1
Greece: income poverty	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6
EU median: income poverty	16.3	16.5	15.9	16.4	15.4	16.1	15.7	15.6	15.4	15.5

Series	Where countries stood in 2024	
	Income poverty (%)	Reported hardship (%)
Czechia	9.5	14.2
Belgium	11.4	17.1
Denmark	11.6	12.8
Netherlands	12.1	7.3
Finland	12.6	8.9
Ireland	12.7	16.9
Slovenia	13.2	13.6
Poland	13.8	11.5
Austria	14.3	13.1
Hungary	14.3	19.8
Slovakia	14.5	28.7
Cyprus	14.6	20.8
Sweden	14.8	9.9
Germany	15.5	7.3
France	15.9	21.8
Portugal	16.6	21.8
Malta	16.9	17.3
Luxembourg	18.1	8.5
Italy	18.9	18.7
Romania	19.0	23.8
Greece	19.6	66.7
Spain	19.7	21.9
Estonia	20.2	9.9
Croatia	20.3	19.8
Lithuania	21.5	11.4
Latvia	21.6	23.3
Bulgaria	21.7	37.4

[Figure 1's first tab](#), “How Greece's hardship gap developed”, shows why. Since 2015, Greece's reported hardship has never dropped below two-thirds of households — it opens near 78% and eases only gradually. Its income-poverty rate spends the same decade hovering near a fifth, barely moving at all. Both EU medians sit far beneath Greece's line for reported hardship, the whole way through. This isn't a single bad year showing up once. It's been the shape of an entire decade.

Switch to the [second tab](#), “Where countries stood in 2024”, and the gap turns from a Greek pattern into a European outlier. Each grey dot is an EU country. The horizontal axis is income poverty; the vertical is reported hardship. The dotted line is the relationship the other twenty-six countries actually follow. Greece is the blue point far above it. At Greece's income-poverty rate, that relationship predicts about 20% of households struggling to make ends meet. Greece reports about 67%. The 47-point distance between those two numbers is the gap this piece tries to understand.

Greek subjective hardship runs 52.6 points above relative income poverty on average, and Greece ranks first of 27 on hardship while ranking seventh on AROP.

The two numbers are not rival opinions. They come from the same European system, from surveys of the same households, but they ask different questions. Income poverty asks whether a household sits below 60% of its country's median income, a measure of relative position. Reported hardship asks something more direct: can the household make ends meet? In most of Europe, those two answers move together. In Greece, they have stayed far apart, year after year.

But once a gap like that appears, the two numbers are not usually read with equal trust. Income poverty feels firmer: a number built from income, thresholds and ranks. Reported hardship feels softer: not necessarily a fact about a household's circumstances, but the way that household describes itself. Is that fair? Or is the hardship signal telling us something material that income poverty alone cannot see?

This Was Not Just a Feeling

If Greek households are simply answering a subjective question more darkly than everyone else, then this is a story about how people talk about their lives, not about poverty. Everything the rest of this piece does next would be measuring an echo, not an economy. That has to be settled first, and settled seriously, not waved off as an obvious no.

The test isn't whether a household *feels* like it's struggling. It's whether that feeling moves together with things that name actual events, not moods: falling behind on bills, being unable to cover a surprise expense, being unable to heat the home properly, going without several ordinary things at once. A vague question can be coloured by a bad mood. Four specific, concrete facts agreeing with it at once is a much harder thing for mood alone to produce.

FIGURE 2 Hardship Moves With Concrete Financial Strain

Did concrete affordability difficulties rise and fall with what Greek households reported?

▼ Methods and limits

Read with this. Same-survey corroboration, not independent validation or causal evidence. Greek correlations with reported hardship: unexpected expenses 0.92 · material deprivation 0.94 · keeping the home warm 0.87 · falling behind on bills 0.37. Every line is a distance from its own 2015-2024 average, including the European median, so three series on different scales can share one axis: the shapes may be compared, the levels may not. The scale is the same in all four tabs. European median hardship is deliberately absent, since the question here is whether the concrete difficulty moved with Greek reports, not how Greece compares. Same-survey corroboration: not independent validation, and not causal evidence.

Unexpected expenses										
Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	5.7	4.8	5.2	2.1	-1.0	-2.2	-1.0	-3.6	-5.0	-5.3
Greece: unexpected expenses	4.7	4.9	4.0	1.7	-0.9	2.0	-2.4	-5.1	-4.4	-4.8
EU median: unexpected expenses	7.5	5.6	2.5	-0.2	-0.9	-1.8	-4.7	-1.1	-2.9	-3.6
Material deprivation										
Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	5.7	4.8	5.2	2.1	-1.0	-2.2	-1.0	-3.6	-5.0	-5.3
Greece: material deprivation	2.0	2.8	2.7	0.5	0.2	-0.7	-1.7	-1.7	-2.1	-1.6
EU median: material deprivation	2.3	1.2	0.8	-0.1	-0.5	-1.0	-0.7	-0.6	-0.6	-1.2
Keeping the home warm										
Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	5.7	4.8	5.2	2.1	-1.0	-2.2	-1.0	-3.6	-5.0	-5.3
Greece: keeping the home warm	7.6	7.5	4.1	1.1	-3.7	-4.5	-4.1	-2.9	-2.4	-2.6
EU median: keeping the home warm	1.5	-0.1	-0.2	-0.9	-0.6	-0.3	-1.1	0.8	1.1	-0.0
Falling behind on bills										
Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	5.7	4.8	5.2	2.1	-1.0	-2.2	-1.0	-3.6	-5.0	-5.3
Greece: falling behind on bills	5.8	4.4	1.4	-0.5	-2.1	-6.6	-7.1	2.0	3.8	-0.7
EU median: falling behind on bills	1.9	2.2	0.3	0.9	-0.1	-1.3	-1.2	-1.5	-0.5	-0.9

[Figure 2](#): three of the four tabs tell the same story. In Greece's own year-by-year data, reported hardship tracks an inability to cover a surprise expense almost exactly ([Unexpected expenses](#)) — a correlation of 0.92. It tracks going without several ordinary things at once even more closely ([Material deprivation](#)), at 0.94. It tracks struggling to keep the home warm at 0.87 ([Keeping the home warm](#)). When one line moves, the others move with it, year after year. And this isn't a Greek peculiarity: pooled across all twenty-seven EU countries, once each country's own average is set aside, the same relationships hold, at 0.63 to 0.80.

The fourth tab ([Falling behind on bills](#)) tells a different story, and it's worth sitting with why. Falling behind on bills — the one item that sounds like the hardest, most factual anchor of the four — tracks reported hardship at only 0.37, by far the weakest of the set. Falling behind requires having had credit and obligations to fall behind on in the first place. A household that lost access to credit years ago, or never had any to lose, can be in real difficulty without that difficulty ever showing up as an unpaid bill. The weakest link in the evidence turns out to have an ordinary explanation, not a suspicious one.

None of this is independent proof, and that has to be said plainly rather than buried. Every one of these items comes from the same interview as the question about making ends meet — the same

household, the same sitting. A household in a genuinely grim mood could rate the whole set grimly, and that alone would produce numbers like these. What's been shown is that reported hardship is coherent with concrete circumstances. It hasn't been shown, and can't be shown this way, that it's confirmed by something entirely outside the survey.

There is a second, different kind of check, and it does not depend on the same hardship questions. If this one number were an isolated fluke — one instrument twitching on its own — it should stand apart from everything else describing Greek life. It doesn't.

FIGURE 3 The Problem Spread Across the Dashboard

Did Greek disadvantage deepen on a few measures, or spread across many?

▼ Methods and limits

Read with this. Descriptive summary of the condition, not a predictor of it. DESCRIPTIVE ONLY. Breadth was tested as a predictor of reported hardship in P3a and does not survive: on its own it is not significant ($p = 0.12$), and once the other accumulated measures enter the model its coefficient reverses sign. It summarises the condition rather than explaining it. THE BASKET IS FIXED: these are the 16 indicators with a valid EU position in both 2008 and 2024, chosen without reference to whether they improved or deteriorated, so the same things are counted at both ends and in every year between. Unemployment, youth unemployment and the employment rate are absent because comparable EU coverage for them begins in 2009; hours worked, pay per hour, the work-effort squeeze and real wages are all present, so this is not a basket without labour-market information. The outcome and every model covariate are excluded. THE EU-COUNTRY MEDIAN is the median of the 27 member states' own shares on this basket -- a median across countries, not Eurostat's population-weighted EU aggregate -- named accordingly rather than simply 'EU'. Croatia never reaches full 16-indicator coverage in any basket year and does not appear as a line for that reason. The 'Which measures' view's axis is POSITION, not value: the indicators have no common unit, so 0 is the best place in the Union to be on that indicator and 100 the worst, whichever direction 'worse' runs in.

Of 16 indicators, **4 were already in the EU's worst fifth in 2008, 7 entered it by 2024, and none left.**

		Which measures																
Series	2008 position					2024 position							Transition					
Hours worked against hourly pay	59.3					100.0							entered worst fifth					
Prices measured against wages	48.1					100.0							entered worst fifth					
Pay per hour worked	55.6					100.0							entered worst fifth					
Household income after inflation	51.9					100.0							entered worst fifth					
Wages after inflation	51.9					100.0							entered worst fifth					
The poverty line after inflation	51.9					100.0							entered worst fifth					
Keeping the home warm	73.1					98.1							entered worst fifth					
Hours worked each week	94.4					100.0							already worst fifth					
What households expect of the year ahead	100.0					100.0							already worst fifth					
Citizens leaving the country	86.4					96.2							already worst fifth					
Income inequality	80.8					81.5							already worst fifth					
Food prices	14.8					77.8							outside both years					
Economic output per person	55.6					77.8							outside both years					
Prices overall	53.7					70.4							outside both years					
Housing and energy prices	63.0					59.3							outside both years					
What households actually spend	48.1					48.1							outside both years					
How many measures																		
Series	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	
Greece	25.0	18.8	43.8	50.0	50.0	50.0	43.8	50.0	50.0	62.5	56.2	56.2	56.2	56.2	56.2	62.5	68.8	
EU-country median	6.2	12.5	9.4	12.5	18.8	18.8	15.6	12.5	18.8	18.8	12.5	12.5	18.8	12.5	12.5	18.8	12.5	

Take sixteen separate measures of Greek life (see [Figure 3](#)) — wages, hours worked, prices, saving, what households expect of the coming year, how many people are leaving — chosen only because both a 2008 and a 2024 reading exist for each, so the count can't grow just because more things got measured later. The first tab ([Which measures](#)) lays out where Greece sat on each one, in each year, and whether it crossed into the EU's worst fifth or was already there. The point isn't that every one of them moved identically — it's that disadvantage spread across the dashboard, not just in one place.

Switch to the second tab ([How many measures](#)) and that spread turns into a single line. Before the crisis, Greece sat in the worst fifth of the EU on four of the sixteen — a quarter. Now it sits there on eleven — two thirds of the whole set. Plotted alongside it, the EU-country median barely moves over the same seventeen years. This isn't a general European drift. Most of these measures don't come from the same interview as the hardship question, either.

One more thing is worth mentioning here, carefully. Put the closest hardship items into the same statistical picture as the official poverty rate, and much of Greece's excess stops standing apart. That matters. But it also carries the same-survey problem all over again, in a sharper form — and it

returns later in this piece, because it turns out to be exactly the point where the explanation looks strongest and least independent.

So the hardship signal is too materially connected to dismiss as mood. It is also too close to the same survey instrument to call independently proven. Both of those are true at once, and the honest version of this section holds them together rather than picking one. What it does settle is enough to move forward on: if the difficulty is real, why doesn't the official poverty rate register it?

▼ The precise numbers behind this section

Reported difficulty co-moves with arrears, inability to meet an unexpected expense, inadequate heating and severe deprivation at within-country correlations of 0.63 to 0.80.

The limits of this. same-instrument corroboration, NOT independent validation | all items and the outcome come from EU-SILC | not uniform: arrears within Greece is 0.371.

Those same four items statistically absorb 71% of Greece's baseline residual (+46.92 to +13.74).

The limits of this. ABSORPTION, never explanation | shared instrument is not shared cause | diagnostic only; never a headline explanation.

The Poverty Line Fell With the Country

The official poverty line isn't fixed. It moves with the very economy it is supposed to be measuring.

The EU's headline poverty measure, at-risk-of-poverty, is not an income level. It is a percentage — 60% of whatever the national median income happens to be, *that year*. In an ordinary economy, where incomes drift up slowly and roughly together, that is a reasonable way to define being poor relative to your neighbours.

Greece's economy, from 2010, was not ordinary. Incomes fell together, hard and fast, and when the median falls, the poverty line falls with it. A household earning exactly what it earned five years earlier could find itself reclassified from poor to not poor — not because anything in its life had improved, but because the ruler measuring it had shrunk to match the collapse around it.

So the official rate barely moved through the worst years. Hold the line where it stood in 2008 instead, adjusted only for inflation, and measured poverty roughly doubles: from under 20% before the crisis to a peak above 40% in 2014.

FIGURE 4 The Poverty Line Looked Stable. Its Value Did Not.

What happened to the line Greek poverty is measured against, and what does a fixed line show instead?

▼ **Methods and limits**

Read with this. The anchored series is an approximation built in this project and is labelled as such wherever it appears. The first view counts PEOPLE, the second counts EUROS. In the second, both lines are the same official threshold: one as published in each year's own money, the other converted into what it could buy in 2008. The cash line recovers and the purchasing-power line does not, and that difference is why the first view's two counts diverge. Both views are Greece only and support no cross-country statement.

Who falls below a fixed line																						
Series	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: fixed 2008 threshold	25.6	22.8	20.7	21.5	21.2	19.7	17.0	18.0	24.6	33.3	39.7	40.6	39.8	39.8	38.3	37.3	34.6	30.8	32.7	33.6	32.0	30.5
Greece: current-year threshold	20.7	19.9	19.6	20.5	20.3	20.1	19.7	20.1	21.4	23.1	23.1	22.1	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6

What the line itself is worth																						
Series	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Threshold in cash terms	4923	5306	5650	5910	6120	6480	6897	7178	6591	5708	5023	4608	4512	4500	4560	4718	4917	5269	5251	5712	6030	6510
Same threshold in 2008 purchasing power	5819	6089	6264	6343	6377	6480	6808	6768	6027	5168	4589	4270	4228	4216	4226	4338	4498	4884	4838	4815	4878	5113

That gap between the moving line and the fixed one is not this piece's own estimate. It is Eurostat's own hardship figure, run backward past 2010 using a method checked against the official series everywhere the two overlap.

▼ **The precise result**

The outcome is Eurostat's official subjective-hardship indicator, extended backwards before 2010 using a constructed series validated against it on 432 overlapping country-years.

The limits of this. pre-2010 provenance is ours, not Eurostat's.

This is a measure of Greece against its own past, and it contains no other country: it cannot show that Greece's line fell further than anyone else's. Nor is the fixed line the “correct” one — the relative measure is doing exactly what it was designed to do, measuring position. When a whole country falls together, those two questions come apart, and that separation is the point.

Europe already knows income alone is too narrow, for exactly this reason. Its broader measure, AROPE, casts a wider net: a household counts if its income is low, or it can't afford a list of ordinary things, or the adults in it are barely working. If [the puzzle above](#) were simply that the income measure is too narrow, this wider one should mostly dissolve it.

It doesn't. AROPE closes a fifth of the gap, and a shrinking fifth at that.

▼ The precise result

Switching to AROPE closes 9.8 of those 52.6 points (19%), leaving 42.8 unexplained, and its contribution shrinks from 11.0 points in 2015 to 7.3 in 2024.

It helps, and it isn't enough. The wider net picks up under a quarter of the distance, and its contribution is shrinking — from eleven points at the start of the decade to seven by the end. As an explanation of the gap, it is weakening rather than strengthening.

Underneath both headline rates sits a further complication: national averages are averages of people, and Greek age groups did not move together through the crisis. Some improved while others did not, and the combined figure sat placidly between them — which matters, because the household answering the survey question is not answering about the national average. It is answering about itself.

Some Parts Recovered.

Others Fell Further Behind

Greece's recovery is real. It just didn't arrive in every part of a household's life at the same time, or at the same speed — and averages hide that as easily as they hide anything else in this story.

Start with what genuinely improved. Long-term unemployment — out of work for twelve months or more, a different and slower-moving thing than the headline rate — stood at 16.4% of the labour force in 2015. By 2024 it had fallen to 5.4%. That is substantial, real progress. Housing-cost pressure eased too. Time does something specific to a household: savings go first, then whatever can

be sold, then the goodwill of relatives, then the ability to absorb any surprise at all. A shorter spell of joblessness is a genuinely different, less corrosive thing than a long one.

Now the part that didn't keep pace. Greek real wages stood at about 77% of their 2008 level in 2015; by 2024, about 68%. Not recovering slowly — not recovering. Greek wages have now been below their pre-crisis level for fifteen consecutive years, longer than any EU country except Hungary. What people can actually buy for that money — consumption adjusted for local prices — did rise substantially, from roughly 14,800 to 21,300 in the units used for these comparisons. That is real, and worth saying plainly, because a piece about hardship can leave the impression that nothing improved. But the EU median rose faster over the same years, so the distance between Greece and its neighbours widened even as Greece's own number climbed.

FIGURE 5 The Recovery Was Uneven

Which measures converged toward the EU and which diverged?

▼ Methods and limits

Read with this. A NARROWING GAP DOES NOT MEAN GREECE IMPROVED. A gap can close because Greece caught up, because the rest of the EU deteriorated, or because everyone moved in the same direction at different speeds. This chart measures RELATIVE CONVERGENCE, not national recovery, which is why both endpoints for Greece and for the EU median are in the tooltip and the table. "71% of the gap closed" and "conditions improved by 71%" are entirely different claims. The plotted quantity is dimensionless because the underlying gaps are measured in percentage points, index points and PPS per head and cannot share an axis.

Measure	Share of 2015 gap closed	Greece 2015	Greece 2024	EU median 2015	EU median 2024	Gap 2015	Gap 2024
Long-term unemployment	0.71	16.40	5.40	3.60	1.70	12.80	3.70
Share below own GDP peak	0.55	24.84	11.20	0.00	0.00	24.84	11.20
Housing-cost overburden	0.40	45.50	28.90	8.70	6.70	36.80	22.20
AROPE	0.26	32.40	26.90	22.50	19.60	9.90	7.30
Income poverty (AROP)	0.20	21.40	19.60	16.30	15.50	5.10	4.10
Real poverty threshold	0.02	65.14	78.77	106.45	119.11	-41.31	-40.34
Material deprivation	0.01	17.60	14.00	7.80	4.30	9.80	9.70
Reported hardship	-0.02	77.70	66.70	28.90	17.10	48.80	49.60
Real household income	-0.06	67.79	84.13	102.25	120.59	-34.46	-36.46
Hardship gap against income poverty	-0.06	56.30	47.10	13.10	1.20	43.20	45.90
Hardship gap against AROPE	-0.08	45.30	39.80	6.40	-2.30	38.90	42.10
Real wages	-0.78	76.91	68.18	101.43	111.85	-24.52	-43.67
Material resources	-0.93	14769.70	21309.90	16230.40	24129.10	-1460.70	-2819.20
Wage-adjusted affordability	-2.55	141.89	173.09	127.22	120.97	14.67	52.12

The shape is consistent: gaps that were mostly about jobs and housing narrowed considerably; gaps that were about wages, resources and what money can buy narrowed barely at all, or widened. But a chart built to fit fourteen measures onto one axis has to abstract away units, so here is the same story in the numbers each measure actually reports in — and the plainest reading of each.

MEASURE	GREECE, 2015→2024	EU MEDIAN, 2015→2024	WHAT ACTUALLY HAPPENED
Long-term unemployment	16.4% → 5.4%	3.6% → 1.7%	Large improvement; Greece remained above the
Housing-cost overburden	45.5% → 28.9%	8.7% → 6.7%	Improved, but the remaining disadvantage was
Real wages, 2008 = 100	76.9 → 68.2	101.4 → 111.8	Greece fell further behind
Material resources	14,770 → 21,310 PPS	16,230 → 24,129 PPS	Resources rose in Greece, but more slowly than
Wage-adjusted affordability	141.9 → 173.1	127.2 → 121.0	Affordability pressure deteriorated sharply relat

Two things in that table are easy to misread, and both matter. An improving number is not the same as a closing gap: material resources rose by nearly half in Greece and still fell further behind, because the EU median rose faster. And whether a gap narrows because Greece caught up, because the rest of Europe slowed down, or both, is not something either version of this comparison can tell you. Convergence is context here, not a mechanism. It describes what happened, not why.

It isn't only pay, either. In the EU's most recent data, Greece is the worst country in the Union for people who needed medical care and didn't get it, because of cost, waiting time or distance — 12.1% in 2024, against a European median of 1.9%, and second-worst in three of the previous four rounds. That is not a statistic this piece can use to explain the hardship gap; nothing here tests it against the other findings. It is simply what “recovery” can coexist with, on the ground, in the same years the headline numbers were improving.

Two of the measures behind this uneven picture also do something more than describe: how much long-term unemployment a country carries, and how much its households can actually afford, each still predict reported hardship once the official poverty rate is already accounted for.

▼ The precise results

Labour-market exclusion predicts hardship beyond AROP and year effects (coef +4.3402, wild-cluster bootstrap p=0.0085).

The limits of this. cross-country association | no causal claim.

Material resources predicts hardship beyond AROP and year effects (coef -0.0013, wild-cluster bootstrap p=0.0055).

The limits of this. cross-country association | no causal claim.

A country can score badly on affordability two ways: by being expensive, or by paying poorly. Greece does both at once, and a household experiencing it does not much care which half is responsible.

So does the third: what a Greek wage is actually worth against Greek prices.

▼ The precise result

Wage-adjusted affordability predicts hardship beyond AROP and year effects (coef +0.3110, wild-cluster bootstrap $p=0.0005$).

The limits of this. cross-country association | no causal claim.

“The unemployment rate came back. The paycheck did not.”

DESCRIPTIVE CORROBORATION

INDEPENDENT DESCRIPTIVE CORROBORATION (GREECE IN FIGURES)

An outside analysis (Greece in Figures, working from newer Eurostat and ELSTAT numbers than the ones used here) lands on the same shape from a different direction and without running any of the tests in this report: Greece is not last in the EU on what people actually consume, but Greek employees put in some of the longest hours in the Union for comparatively low pay per hour, and everyday things like food cost more than that pay would suggest. Same picture, different route, no formal test behind it.

What this lets us say. The independent picture -- Greece not last on actual consumption, long hours for comparatively low hourly reward, high relative prices in food and information/communication -- is consistent with and external corroboration for this project's own material-resources and wage-adjusted-affordability findings. Its central descriptive numbers reproduce against the primary Eurostat and ELSTAT releases they draw on.

What it does not. Treating the article's own comparisons, or its constructed AIC-per-hour ratio, as inferential evidence: it applies no bootstrap, no FDR correction, no within/between decomposition and no model comparison, and it does not test whether these factors account for reported hardship. Merging its 2025/2026-vintage figures into this project's frozen 2015–2024 panel. Repeating its car-ownership sentence, which cites a vehicle-STOCK figure as evidence of new purchases, or its 'all Greeks travelled' generalisation, which the underlying ELSTAT survey does not support. Reading its AIC rank, hours rank or price-level figure as contradicting this project's own numbers without first noting the different vintage or aggregate behind each.

Greece in Figures, "Γιατί οι Έλληνες νιώθουν τόσο φτωχοί" [Why Greeks feel so poor]. [source](#)

A Decade of Damage Still Counts

Two countries can look identical today and have arrived by different roads. One has had high long-term unemployment for a year. The other has had it for twelve. The intuition that these are not the same situation is strong, and it is the intuition behind almost every account of the Greek crisis.

Intuition isn't evidence, and this is where an appealing story is easiest to oversell. Three earlier drafts of this piece oversold it, in fact, before the wording below was settled.

For each present-day measure in the last section, there is a matching one that counts not the level today but the total weight a country has carried since before the crisis: how much excess unemployment it accumulated, how many consecutive years its wages stayed below 2008, how much its housing costs deteriorated.

FIGURE 6 Greece Had Among Europe's Largest Accumulated Burdens

How much accumulated unemployment has each country absorbed, and where does Greece sit?

▼ Methods and limits

Read with this. cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out The three measures are in different units and may not be compared with each other: accumulated unemployment and housing deterioration are percentage-point-years, wage non-recovery is a count of consecutive years.

Accumulated unemployment

Country	Accumulated unemployment (percentage-point-years above 2009)
Greece	137.5
Cyprus	64.1
Croatia	43.5
Spain	33.1
Italy	32.9
Bulgaria	26.6
Portugal	20.6
Slovenia	20.6
Netherlands	13.1
Luxembourg	9.1
Ireland	8.9
Slovakia	8.9
Poland	7.7
France	5.9
Lithuania	5.6
Denmark	5.6
Hungary	3.2
Estonia	3.1
Finland	2.9
Austria	2.5
Belgium	2.4
Romania	2.4
Latvia	2.0
Czechia	1.2
Sweden	0.6
Malta	0.0
Germany	0.0

Years wages below 2008	
Country	Years wages below 2008 (consecutive years below the 2008 level)
Hungary	16.0
Greece	15.0
Italy	14.0
Finland	3.0
Sweden	2.0
Slovenia	0.0
Romania	0.0
Portugal	0.0
Poland	0.0
Netherlands	0.0
Malta	0.0
Latvia	0.0
Luxembourg	0.0
Lithuania	0.0
Austria	0.0
Ireland	0.0
Belgium	0.0
Croatia	0.0
France	0.0
Spain	0.0
Estonia	0.0
Denmark	0.0
Germany	0.0
Czechia	0.0
Cyprus	0.0
Bulgaria	0.0
Slovakia	0.0

Country	Housing deterioration since 2010
	Housing deterioration since 2010 (percentage-point-years above 2010)
Greece	233.2
Bulgaria	116.3
Luxembourg	37.8
Portugal	35.8
Sweden	17.1
Estonia	11.8
Slovenia	9.9
Hungary	9.6
Germany	9.4
Italy	8.2
Belgium	7.9
Finland	7.0
France	6.6
Slovakia	6.1
Latvia	5.7
Netherlands	4.9
Ireland	4.6
Spain	4.5
Malta	4.5
Austria	4.3
Poland	4.2
Romania	4.2
Czechia	3.8
Cyprus	2.1
Lithuania	0.5
Croatia	0.0
Denmark	0.0

Greece carries more of this weight than almost anywhere else in Europe. On one of the three measures it is not the heaviest — Hungary has a longer run of depressed wages — and the chart shows every country rather than Greece alone, so that comparison is visible rather than buried.

Carrying a heavy history is one thing. Whether that history still matters once you already know where a country stands today is a harder question, and a more important one: if you know this year's unemployment rate, does the decade behind it tell you anything more?

For three measures, it does.

How much excess unemployment a country has absorbed since before the crisis. How many years its wages have run below their 2008 level, on one specific way of counting that — other reasonable ways of counting the same idea point the same direction without quite clearing the bar, worth knowing even though they don't change the answer. And, more tentatively, how much its housing costs have deteriorated.

▼ The precise results

Accumulated excess unemployment predicts hardship (bootstrap $p=0.0025$) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap $p=0.0025$).

The limits of this. cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out.

Duration of real wages below their 2008 level predicts hardship (bootstrap $p=0.0245$) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap $p=0.0060$).

The limits of this. cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out | supported only as the CURRENT uninterrupted run against a fixed 2008 base; alternative constructions point the same way but do not meet the criteria.

Housing-cost deterioration since 2010 predicts hardship (bootstrap $p=0.0460$) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap $p=0.0015$).

The limits of this. cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out | BORDERLINE: bootstrap $p=0.0460$, 91 of 1,999 exceedances.

The housing result is the shakiest of the three, and it is presented that way rather than rounded up to a clean yes.

The pattern is not universal, though, which is itself informative.

For the cost-of-living measure, it runs the other way: today's number carries the signal, and the accumulated version of it doesn't resolve.

▼ The precise result

For wage-adjusted affordability the pattern reverses: the current measure survives conditioning (bootstrap $p=0.0035$) while the accumulated one remains inconclusive.

The limits of this. the accumulated measure is inconclusive, NOT unsupported.

If history mattered as a general rule, that reversal shouldn't happen. It suggests these results are specific to work, wages and housing rather than some broad law that the past always counts for everything.

One more piece of the history simply isn't there to look at. What households could actually afford, tracked back to before the crisis, can't be built at all — the source data only starts in 2015, already years into the recovery. Moving the starting line earlier would have solved the data problem and created a different one: a measure that could no longer see the crisis it exists to describe. So it stays out, and is named here rather than quietly absent.

▼ The precise result

Accumulated material resources (C1) could not be tested at all: the source series begins in 2015 and no 2008 baseline exists. The baseline was not moved to make it testable.

The limits of this. infeasible, not null.

The accumulated wage-shortfall coefficient cleared every conditional gate but its prior stage did not support it, so the pre-registered ceiling caps it. It is reported and is not a finding.

The limits of this. capped by the E7 ceiling: E7 may only qualify or withdraw.

Wages are the exception worth naming for the opposite reason. The accumulated version of the wage story held up under every check put in front of it — as convincingly as the three results above. It isn't counted among them, because the present-day wage measure underneath it was never established as real in its own right, earlier in this piece. A result can only stand as high as its foundation, and this one's foundation was never poured. It's named here, not because it counts, but because pretending it doesn't exist would be its own kind of dishonesty.

Where the Evidence Stops

Everything in [the previous section](#) is a statement about how countries differ from each other. It is tempting, and wrong, to turn it into a statement about how Greece changed over time.

Look for that change happening inside Greece itself, over the years, and the evidence isn't there to find it — not because it has been ruled out, but because this kind of comparison cannot see it either

way.

▼ The precise result

No accumulated measure permits dynamic wording. Across P5, E4 and E7 no within-country estimate is significant in the adverse direction and no first-difference test supports one.

The limits of this. three RELATED checks on one panel, not independent replications | the dynamic tests were not multiplicity-corrected.

FIGURE 7 The Evidence Is Mostly Between Countries

Is the accumulated effect between countries, or within one over time?

Accumulated measure	Between (SD)	Within (SD)	Between vs within		First difference (raw)	First difference p
			Between p	Within p		
Accumulated unemployment	0.5924	0.0296	0.0000	0.6867	0.0461	0.7324
Cumulative wage shortfall	0.6584	-0.0544	0.0130	0.3147	-0.0596	0.0284
Years wages below 2008	0.7271	0.0700	0.0025	0.0823	0.0721	0.4952
Cumulative GDP shortfall	0.3890	0.0674	0.0289	0.0600	0.0273	0.1442
Cumulative threshold shortfall	0.8271	0.0123	0.0000	0.5599	-0.0148	0.3917
Accumulated affordability pressure	-0.3953	-0.0764	0.3639	0.2825	-0.0298	0.0000
Compounded inflation since 2008	-0.1718	-0.1228	0.4947	0.3738	-0.0314	0.5766
Housing deterioration since 2010	0.9282	-0.0658	0.0000	0.2343	-0.0090	0.8679

“Countries that accumulated more hardship report more difficulty” is a claim about a group photograph. “As Greece accumulated more hardship, Greek households reported more difficulty” is a claim about a film. This piece has the photograph. It does not have the film — not because the film doesn't exist, but because seeing it would mean giving up the comparison between countries, and what's left on its own isn't sharp enough to say either way.

Not every present-day measure earned a place in the story so far, either. Nine were tried; three worked. The other six mostly went quiet rather than failed outright — with twenty-seven countries and a decade of data, most of them could only have caught an effect bigger than any effect worth caring about. Silence isn't a verdict.

A couple of them can be set aside for real, at least at the size this design could catch — both measures of price inflation among them. The rest simply weren't put under enough pressure to say either way.

▼ The precise results

Six of nine current-level constructs are inconclusive under available power, not unsupported; two cleared FDR and collapsed under the bootstrap ($p=0.40$ and 0.55).

The limits of this. inconclusive is not evidence of absence.

Annual food and housing inflation are unsupported with adequate power at 0.70 SD; annual headline inflation is unsupported at its detectable conditional magnitude; compounded inflation since 2008 remains inconclusive.

The limits of this. the exclusions are narrow and magnitude-specific | compounded inflation is INCONCLUSIVE, not ruled out.

The biggest complication is a genuine reversal, and it concerns the deprivation items [from earlier](#) — can't pay bills, can't heat the home — which absorbed most of Greece's unexplained excess. Those items come from the same survey as the thing they're explaining, and whether to let a measure like that into the picture is a real, defensible choice either way. So both versions were built.

Choose differently, and Greece's whole position in Europe flips.

▼ The precise result

Greece's residual is $+6.93$ (rank 3/27) in the main model and -9.39 (rank 25/27) when the same-instrument deprivation predictor is removed, on identical rows. Neither specification is definitive.

The limits of this. NEITHER specification is definitive | the two may not be merged or averaged | selection between them may not be made on residual size | conclusions about absorption depend materially on whether same-instrument deprivation is admitted.

FIGURE 8 What the Recovery Story Still Misses

Does the answer depend on which model is chosen?

▼ Methods and limits

Read with this. NEITHER specification is definitive. They may not be merged or averaged, and selection may not be made on residual size.

Country	Reference specification	Deprivation-free companion	Change
Luxembourg	8.80	15.48	6.68
Cyprus	7.05	10.75	3.70
Greece	6.93	-9.39	-16.32
Latvia	6.32	4.65	-1.66
Hungary	4.86	7.42	2.56
Bulgaria	4.06	14.65	10.60
Czechia	3.53	-3.63	-7.16
Slovakia	3.28	-0.68	-3.96
Belgium	3.13	5.26	2.13
Ireland	3.05	6.14	3.09
Croatia	2.62	-0.38	-2.99
France	1.85	3.88	2.02
Austria	1.36	0.95	-0.41
Portugal	1.30	0.62	-0.67
Malta	1.28	0.36	-0.91
Poland	0.56	-4.32	-4.88
Denmark	0.16	-3.65	-3.81
Sweden	-1.00	-4.55	-3.55
Slovenia	-2.66	-6.33	-3.67
Netherlands	-3.47	-4.87	-1.39
Lithuania	-4.17	-1.81	2.36
Germany	-4.31	-6.00	-1.69
Estonia	-4.49	-10.12	-5.63
Finland	-5.07	-6.80	-1.72
Italy	-6.35	-5.18	1.17
Romania	-10.32	9.76	20.08
Spain	-10.63	-10.06	0.57

“That is not a wobble. It is a reversal, and there is no honest way to pick between the two.”

Greece moves from the third-worst country in Europe on unexplained hardship to the twenty-fifth — from a stark positive outlier to a stark negative one — on exactly the same rows of data, with one measure added or removed. The two results can't be averaged, and can't be chosen between by which looks more plausible; doing that is exactly what would make a check like this meaningless. Which is why nothing later in this piece leans on it.

Two further ideas were meant to carry real weight here, and neither survived contact with the data — worth naming rather than quietly dropping. A synthetic Greece, built to track the real one before 2008 and read the divergence after as the crisis effect, collapsed into a blend of essentially two countries and missed four of the six conditions it had been required to meet before anyone looked at the result. Its chart is not shown here, because a dramatic picture built on a counterfactual that thin would persuade readers of something the evidence can't actually support. And the sixteen-measure spread [from earlier](#) made Greece's position worse, not better, once it was asked to predict rather than just describe, and its direction flipped once other measures were held steady — left deliberately unexplained here, since inventing a story for a strange result in a design that already failed is how failed ideas come back from the dead.

Both are recorded for what they are: attempts that didn't work, kept visible rather than erased.

▼ The precise results

The synthetic-control comparative design failed four of six pre-registered gates and is not a usable comparison. Its divergence figure is kept out of every document.

The limits of this. donor weights collapse to Hungary 0.55 and Bulgaria 0.45 | the divergence figure is NON-REPORTABLE.

Multi-domain breadth failed the incremental criterion: adding it to the main model worsened Greece's residual from 6.93 to 10.39 and reversed its sign conditionally. The reversal is left uninterpreted.

The limits of this. a result about the specification, not about power | the sign reversal is deliberately left uninterpreted.

What Greece's Recovery Leaves Out

There is a simpler explanation for everything above, and it deserves to be taken seriously rather than waved away: maybe Greeks are just gloomier answerers. The evidence so far can't settle that on its own — it all comes from inside one survey. A better test looks across different subjects entirely. A general tendency to answer darkly should drag everything down about equally; a pattern that's extreme on money and milder elsewhere points at circumstances instead.

MEASURE	GREECE	EU MEDIAN	GREECE'S POSITION
Reported hardship	66.7%	17.1%	1st of 27, worst first
Financial expectations	-43.2	-1.8	1st of 27, worst first
Life satisfaction	6.7	7.3	2nd of 28, worst first

Greece is worst in Europe on the two money questions and close to worst on general life satisfaction — a difference of degree, not of kind, which points toward circumstance rather than temperament, though it doesn't prove it.

▼ The precise result

Greece is consistently among Europe's worst countries on life satisfaction and consistently worst on financial hardship and expectations. The greater extremity of the financial indicators suggests domain specificity, but does not rule out a broader negative reporting tendency.

The limits of this. Greece is second-WORST on life satisfaction by 2024, not middling | a broader negative reporting tendency is NOT ruled out | Greek life satisfaction ROSE over the observed period, 6.2 to 6.9; only its rank worsened, because other countries improved faster | the series begins in 2013, at the crisis trough, with no pre-crisis baseline.

One thing about that life-satisfaction number is worth holding onto, because it's easy to get backwards: it actually *rose* over the period. Greece's rank against it fell anyway, because its faster-improving neighbours pulled further ahead. A falling rank is not the same thing as a falling number.

DESCRIPTIVE CORROBORATION

FINANCIAL EXPECTATIONS AND LIFE SATISFACTION

Greece's money questions sit at the far edge of the European range while its general-wellbeing question sits closer to it. That difference of degree is what the comparison shows, and it is description rather than a test.

What this lets us say. The financial indicators are more extreme than the general-wellbeing one, which suggests domain specificity. A broader negative reporting tendency is NOT ruled out.

What it does not. Describing Greece as ordinary or middling on life satisfaction: it is second-worst in the EU by 2024. And reading a worsening RANK as falling satisfaction, when the Greek level rose over the period.

reporting_style_cross_indicator.csv, this project

One further check reaches back before the Eurostat series begins. A separate European survey shows Greece already sitting about 0.8 points below its comparison group before the crisis, falling further during it, and recovering its level — but not its relative position — by the 2020s. A long-standing pattern of lower reported wellbeing remains plausible on this evidence. It also cannot be established by it.

DESCRIPTIVE CORROBORATION

PRE-CRISIS WELLBEING BASELINE (ESS)

Across six rounds of a separate European survey, holding the same twelve countries fixed each time, Greece's level fell and then recovered while its position relative to the others did not. This is descriptive corroboration and not a test.

What this lets us say. On a balanced set of 12 countries observed in every Greek ESS round, Greece already sat about 0.8 points below the median before the crisis, fell to 5.64 by 2010/11, and recovered its LEVEL to roughly the pre-crisis value by 2023/24. Its comparative position did not recover: the gap to the balanced median is wider than before the crisis, and Greece has been worst of the 12 since 2010/11. A longstanding low-wellbeing pattern therefore remains plausible and generic pessimism or reporting culture CANNOT be dismissed.

What it does not. Splicing ESS to the Eurostat EU-SILC series, or reading the two as one trajectory. Attaching any confidence interval, standard error or significance test to these means, which are approximate reconstructions from displayed weighted percentages. Comparing ALL-COUNTRY ranks across rounds, since the country set varies from 22 to 30. Treating the decade after 2010/11 as observed.

European Social Survey Data Portal, public Analysis tab for stflife (life satisfaction, 0-10), rounds 1, 2, 4, 5, 10 and 11. [source](#)

Several other factors that come up in every account of the Greek crisis were not established here — institutional trust, the design of the adjustment programmes, migration, and the incidence of indirect taxes among them. Leaving them out silently would be misleading; treating them as findings would be worse.

▼ What this piece did not test

CONTEXTUAL EVIDENCE

INSTITUTIONAL TRUST

Trust in institutions is low in Greece, and there is a plausible route by which it could matter: a household that doesn't expect help to arrive may experience the same circumstances as more frightening. This piece has no check on that either way.

What this lets us say. May affect how households interpret insecurity; available evidence does not identify an independent effect.

What it does not. Presenting trust as an explanation of the residual, or implying it was tested and found to matter.

OECD (2024), OECD Survey on Drivers of Trust in Public Institutions - 2024 Results, Country Notes: Greece. OECD Publishing, Paris. [source](#)

LITERATURE-GROUNDED CONTEXT

CRISIS AND ADJUSTMENT POLICIES

The bailout programmes from 2010 reshaped incomes, job protections, pensions and public services at once and in a hurry. They are the backdrop to every accumulated measure described earlier.

What this lets us say. Help explain the historical setting; this project does not estimate their causal contribution.

What it does not. Attributing any share of the accumulated exposure to a specific programme or measure.

Andriopoulou, E., Kanavitsa, E. & Tsakloglou, P. (2020), Decomposing Poverty in Hard Times: Greece 2007-2016. LSE GreeSE Paper No. 149. [source](#)

CONTEXTUAL CONSEQUENCE

MIGRATION

Large numbers of working-age Greeks left during the crisis, and some have returned. This cuts both ways: plausibly a consequence of a broken labour market, and plausibly part of why the people who stayed look the way they do.

What this lets us say. A plausible consequence of prolonged labour-market damage and a possible contributor to it. It is UNSUPPORTED as an independent aggregate predictor here.

What it does not. Reading it as a driver, or as ruled out. E3 tested it on this panel and found nothing ($p=0.4006$), which speaks to aggregate prediction and not to either causal direction.

Lazaretou, S. (2016), The Greek brain drain: the new pattern of Greek emigration during the recent crisis. Economic Bulletin, Bank of Greece, issue 43, pp. 31-53. [source](#)

FUTURE HYPOTHESIS

TAX BURDEN AND UNEQUAL TREATMENT

Published research establishes that Greece's system of indirect taxes became markedly harder on lower-income households across the crisis. If a household's position worsened through tax in a way income-based poverty measures capture badly, that would be one route to the kind of gap this report describes.

What this lets us say. Published incidence evidence establishes that Greece's indirect tax system became markedly more regressive over the crisis. Whether that channel contributes to the hardship gap is a plausible hypothesis and is NOT tested here.

What it does not. Any quantitative statement linking tax burden to the hardship gap. The literature is about incidence, not about this outcome, and this project tested nothing on tax.

Kaplanoglou, G. (2015), Who Pays Indirect Taxes in Greece? From EU Entry to the Fiscal Crisis. Public Finance Review 43(4), 529-556. [source](#)

Greek households report struggling at a rate far above what the official poverty figure predicts, and have done so consistently for a decade. Part of that distance is now easier to understand: official measures are narrower than the experience they are used to summarise; what households report corresponds to real material trouble, even if that agreement comes from inside a single survey; and both the present and the accumulated past carry information the official rate does not.

What the evidence stops short of is just as much part of the answer. Nothing here says what causes what. Nothing here shows Greece changing over time. Most of what didn't work is unresolved, not ruled out. One central result flips on a judgement call the data cannot settle. And a plainer explanation — that Greeks simply answer more darkly — cannot be fully excluded. Most of the 52.6-point gap is still unexplained.

AUTHOR INTERPRETATION

POLICY IMPLICATIONS

What runs through all of this is that no single official number captures what Greek households are reporting. Each one misses something different, and the ones that catch what the others miss are not the headline.

What this lets us say. POLICY RECOMMENDATION, NOT AN EMPIRICAL CONCLUSION: poverty dashboards should combine AROP, its real threshold, anchored poverty, AROPE/deprivation and accumulated labour and housing indicators.

What it does not. Presenting this as a finding. It follows from what the analysis showed AROP alone misses; it is not itself a result.

Greece's recovery was real. So was the hardship that remained. The official poverty rate is not wrong; it answers a narrower question than the one Greek households have been living. A country can recover in its averages while its households continue to carry the history.